

Experiment 32:-

Construct a C program to simulate the Least RecentlyUsed paging technique of memory management.

Code:-

```
#include <stdio.h>

#define MAX_FRAMES 3

void printFrames(int frames[], int n)
{
    for (int i = 0; i < n; i++)
    {
        if (frames[i] == -1)
        {
            printf(" - ");
        }
        else
        {
            printf(" %d ", frames[i]);
        }
    } printf("\n");
}

int main()
{
    int frames[MAX_FRAMES];
    int usageHistory[MAX_FRAMES];
```

```
for (int i = 0; i < MAX_FRAMES; i++)  
{  
    frames[i] = -1;  
    usageHistory[i] = 0;  
}
```

```
int pageFaults = 0;
```

```
int referenceString[] = {  
    7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2  
};
```

```
int n = sizeof(referenceString) / sizeof(referenceString[0]);
```

```
printf("Reference String: ");
```

```
for (int i = 0; i < n; i++)  
{  
    printf("%d ", referenceString[i]);  
}
```

```
printf("\n\n");
```

```
printf("Page Replacement Order:\n");
```

```
for (int i = 0; i < n; i++)  
{  
    int page = referenceString[i];
```

```
int pageFound = 0;
for (int j = 0; j < MAX_FRAMES; j++)
{
    if (frames[j] == page)
    {
        pageFound = 1;
        for (int k = 0; k < MAX_FRAMES; k++)
        {
            if (k != j && frames[k] != -1)
            {
                usageHistory[k]++;
            }
        }
        usageHistory[j] = 0;

        break;
    }
}
if (!pageFound)
{
    printf("Page %d -> ", page);
    int lruPage = 0;
    for (int j = 0; j < MAX_FRAMES; j++)
    {
        if (frames[j] == -1)
        {
```

```

        lruPage = j;
        break;
    }

    if (usageHistory[j] > usageHistory[lruPage])
    {
        lruPage = j;
    }
}

int replacedPage = frames[lruPage];
frames[lruPage] = page;
usageHistory[lruPage] = 0;

if (replacedPage != -1)
{
    printf("Replace %d with %d: ",
           replacedPage, page);
}
else
{
    printf("Load into an empty frame: ");
}

printFramesframes, MAX_FRAMES);
pageFaults++;
}

```

```
}  
  
printf("\nTotal Page Faults: %d\n", pageFaults);  
  
return 0;  
}
```

Output:

```
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2  
  
Page Replacement Order:  
Page 7 -> Load into an empty frame: 7 - -  
Page 0 -> Load into an empty frame: 7 0 -  
Page 1 -> Load into an empty frame: 7 0 1  
Page 2 -> Replace 7 with 2: 2 0 1  
Page 3 -> Replace 2 with 3: 3 0 1  
Page 4 -> Replace 1 with 4: 3 0 4  
Page 2 -> Replace 3 with 2: 2 0 4  
Page 3 -> Replace 2 with 3: 3 0 4  
Page 2 -> Replace 4 with 2: 3 0 2  
  
Total Page Faults: 9  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```